



PSYXX16: Attention Lecture 1

Alex Holcombe

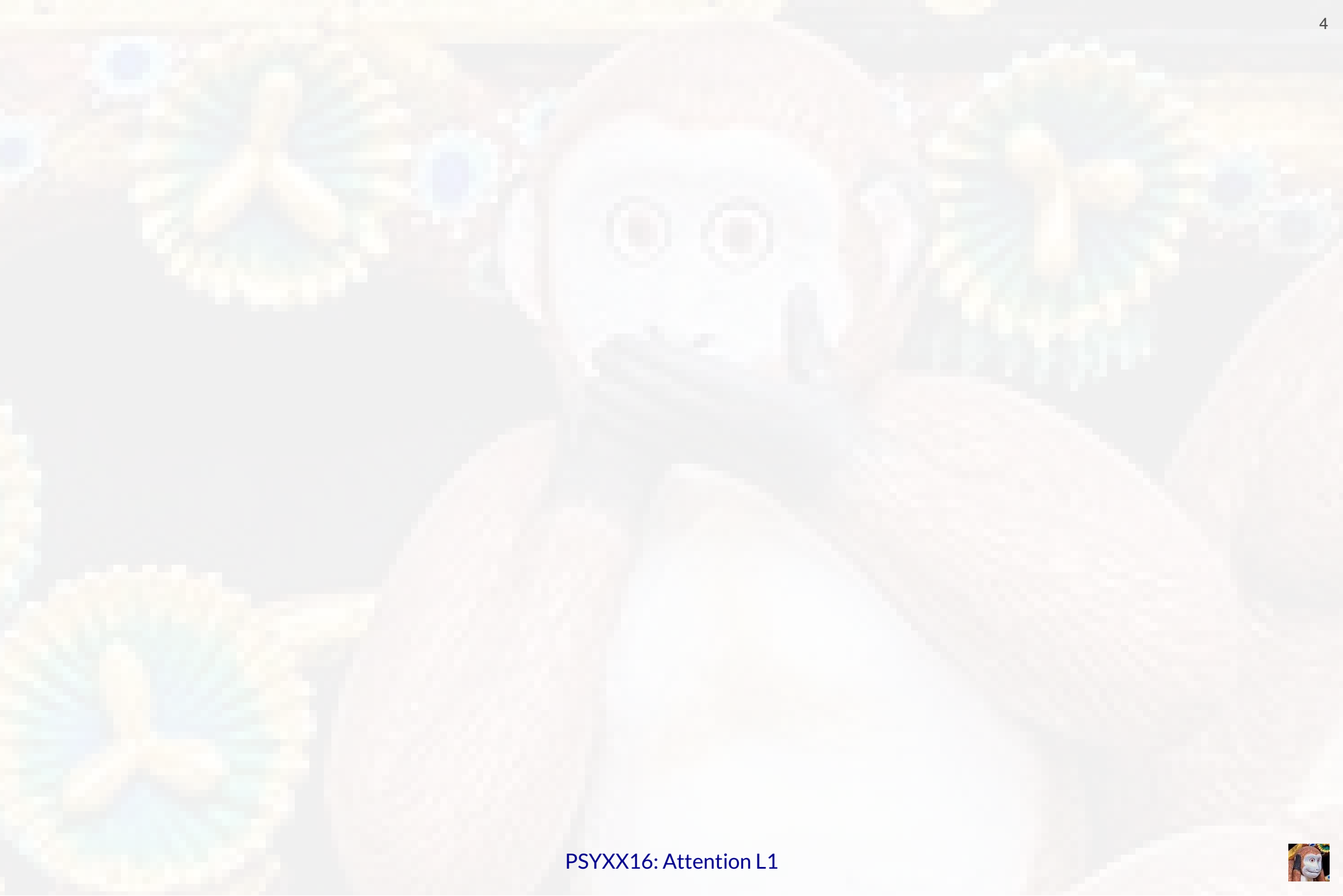
2026-09-01





Alex's journey







All day, the world makes its demands.
There's so much of it, world
begging to be noticed.



All day, the world makes its demands.
There's so much of it, world
begging to be noticed.

– Leila Chatti

- Why do we get distracted?



“Everyone knows what attention is.

It is the taking possession by the mind,
in a clear and vivid form

one out of several simultaneously possible objects or
trains of thought.

Focalization, concentration of consciousness are of its
essence.

It implies withdrawal from some things in order to deal
with others.”

— *William James (1890)*



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- So why does this happen; why do we need attention?
- Why do we get distracted?



Organisation of Alex's material

- The attention textbook: [PSYC2016.whatanimalssee.com](https://psyc2016.whatanimalssee.com)
- These slides: [linked from Canvas](#)
- The Week 6 tutorial: Finding and noticing things in the lab, and in the real world



Frequently asked questions

<https://psyc2016.whatanimalssee.com/index.html>

Can the lecture be downloaded as PDF?

Yes.

->Tools->PDF Export
or press the letter 'e'



What's the point of the lectures if everything is in the textbook?

Lectures:



I can't see everything on some textbook/lecture pages

Some things are offscreen on most phones, but will work on a laptop, desktop, or tablet



TODAY

- Intro: Attention and distraction
- Limits on body
- Limits on mind
- Bottlenecks
- Attentional selection
- Overt versus covert attention
- More about bottlenecks
- Relationship to Consciousness



Understand how unscientific ways of talking about attention relate to scientific concepts (Ch.1,3)

“It implies withdrawal from some things in order to deal with others”

Very inconvenient!



Understand how unscientific ways of talking about attention relate to scientific concepts (Ch.1,3)

Differentiate physical limits from central bottlenecks (Ch.3)

“I can’t be everywhere at once”

- Limit on our abilities, imposed by physics!



Understand how unscientific ways of talking about attention relate to scientific concepts. (Ch.1,3)

Differentiate physical limits from central bottlenecks (Ch.3)

I can't be everywhere at once

I can't do everything at once

- Only have two hands
- Only have one mouth



Differentiate physical limits from central bottlenecks (Ch.3)

Limits imposed by our body

- Can only be in one place at a time
- Can act on only a few things at a time
- Can look directly at only one thing at a time
- *Selection* required

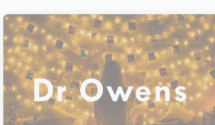


The processing limit is even worse than one might expect from the physical limits

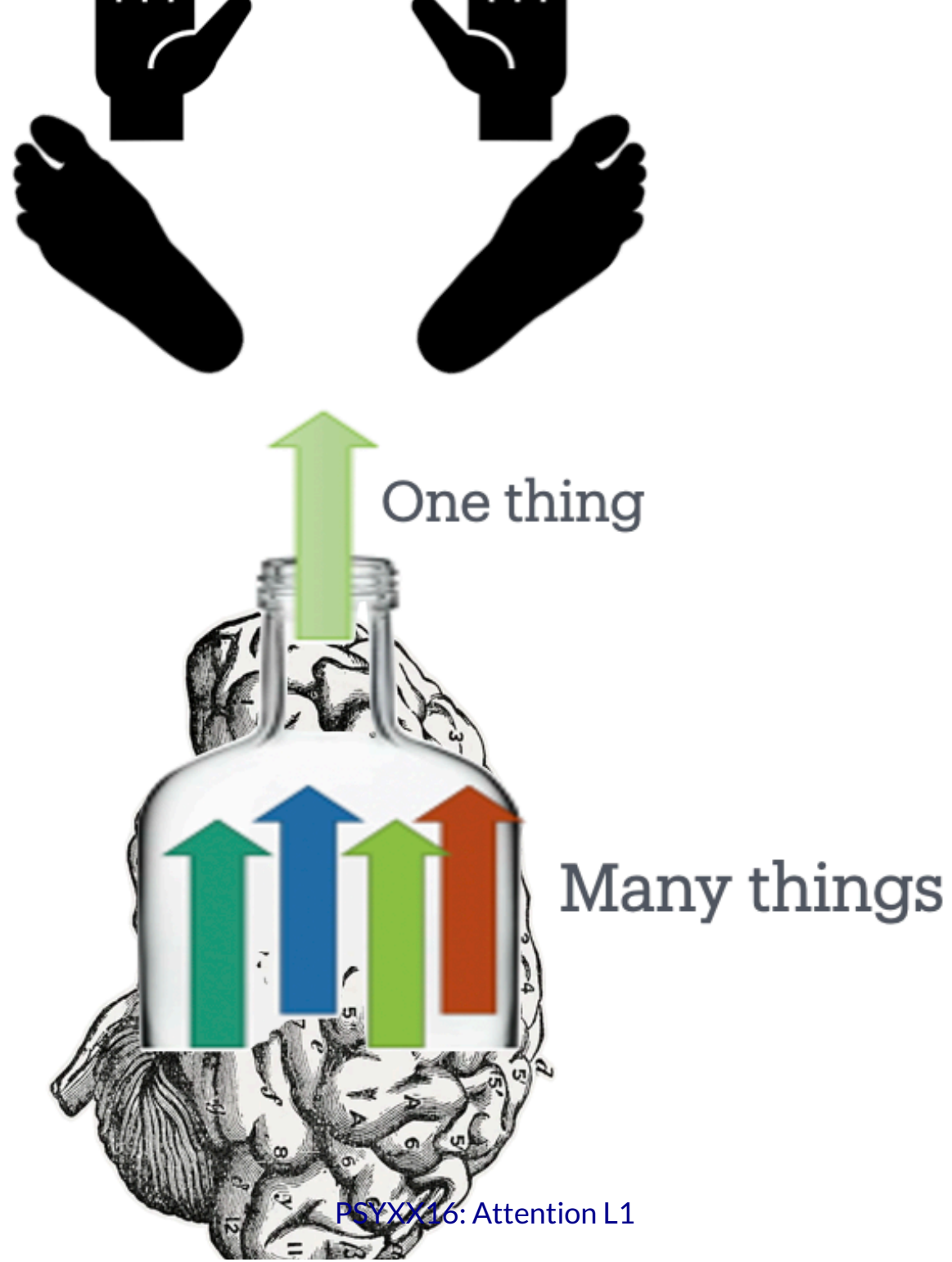
- Can only *extensively mentally process* simultaneously a limited number of things
- “It implies withdrawal from some things in order to deal with others.”



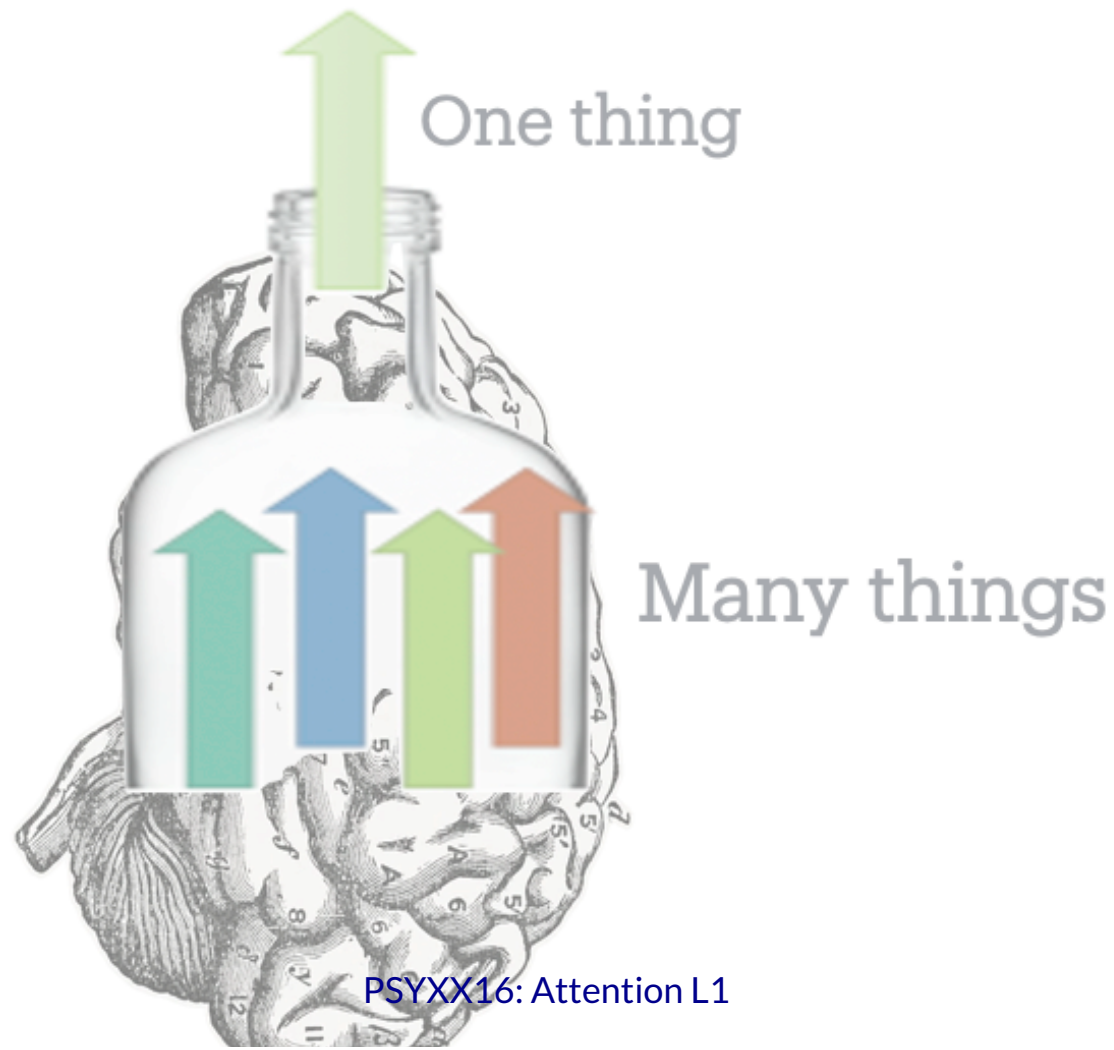


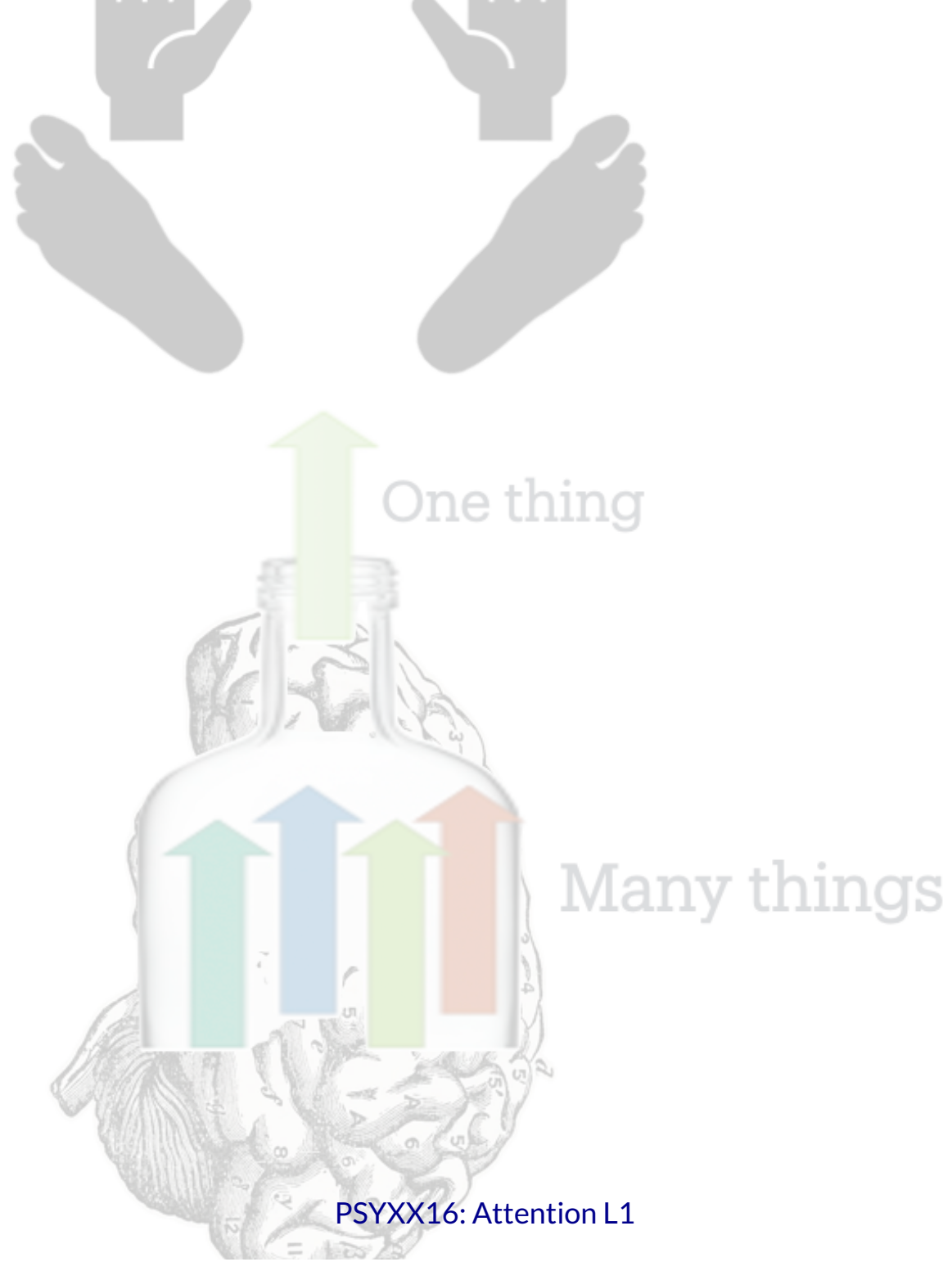
	Week One to Week Four Vision →
	Week Four to Week Five Hearing & Consciousness →
	Week Six to Week Seven Attention →
	Week Seven to Week Nine Theories and Methods in Cognition, Memory Systems and Processes →
	Week Nine to Week Eleven Thinking, Problem Solving, and Decision-Making →
	Week Twelve to Week Thirteen Theories and Research on Intelligence(s) →





Differentiate physical limits from central bottlenecks (Ch.3)





Covert attention

Understand how unscientific ways of talking about attention relate to scientific concepts (Ch.1,3)



- “Sorry, I wasn’t listening!”
 - Were your ears not open?



- “Sorry, I missed that!”
- “I didn’t notice that!”



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Two kinds of selection

Distinguish overt from covert attention (Ch.4)

- Can look directly at only one thing at a time (physical limit)
 - Directing gaze is *overt attention*
- Can only extensively mentally process a few things at once (mental limit)
 - Directing mental resources is *covert attention*



Just two letters can overwhelm capacity

Processing capacity: How many things can be processed at once.

One letter at a time



Just two letters can overwhelm capacity



Z · L

▶ 0:00 / 0:01



Just two letters can overwhelm capacity

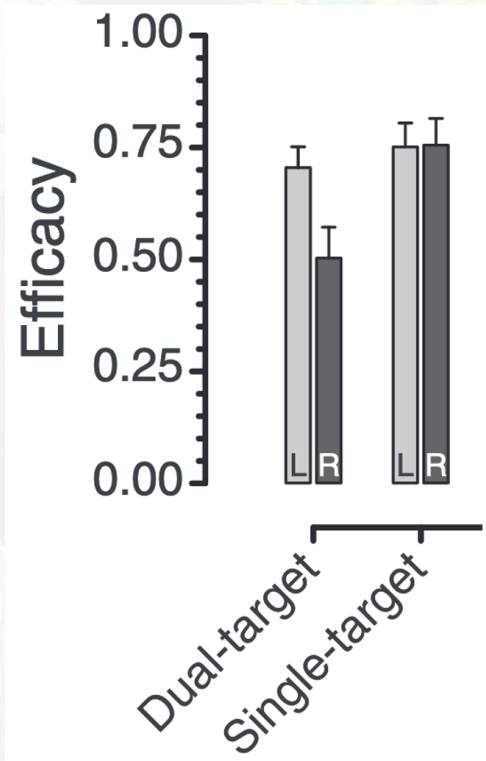


H . A

▶ 0:00 / 0:01



Just two letters can overwhelm capacity



Goodbourn PT, & Holcombe AO (2015)

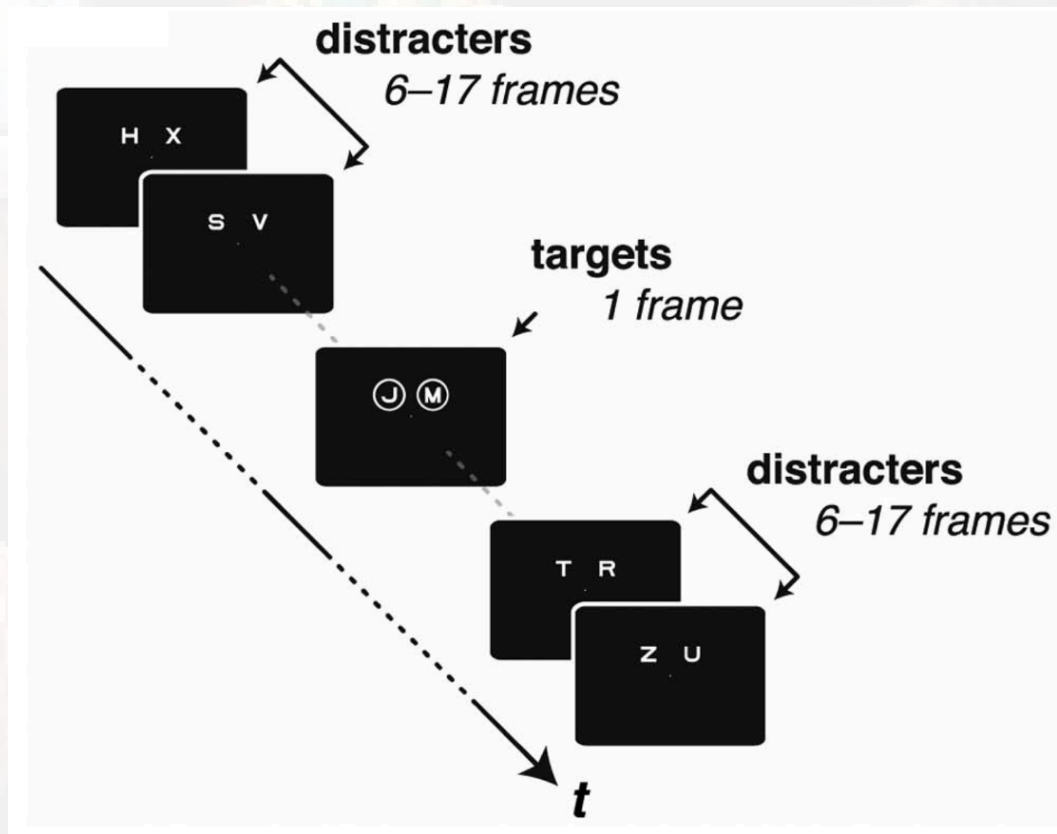


You didn't get both letters right?

- A rate that is slow enough to process one letter is too fast for two
- Selection = choice of which stimulus, or stimuli, to extensively process



Just two letters can overwhelm capacity



To test capacity, we have to prevent use of lingering activation.



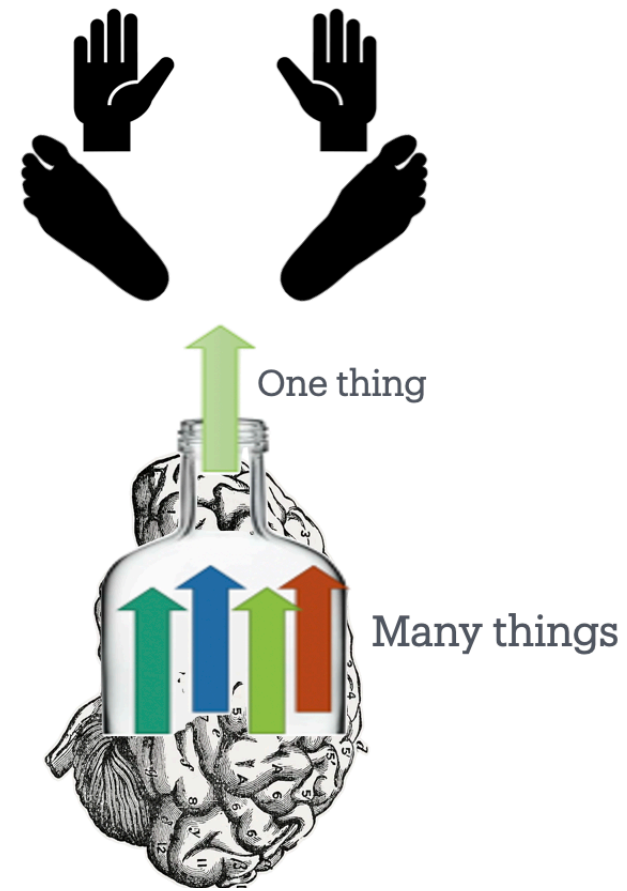
Bottleneck - for letter identification

Differentiate physical limits from mental bottlenecks (Ch.3)



Can only extensively process a few things at once

Limited resources; a bottleneck



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Thing(s) I didn't
understand

Waiting for responses ...



Consciousness

- David Alais

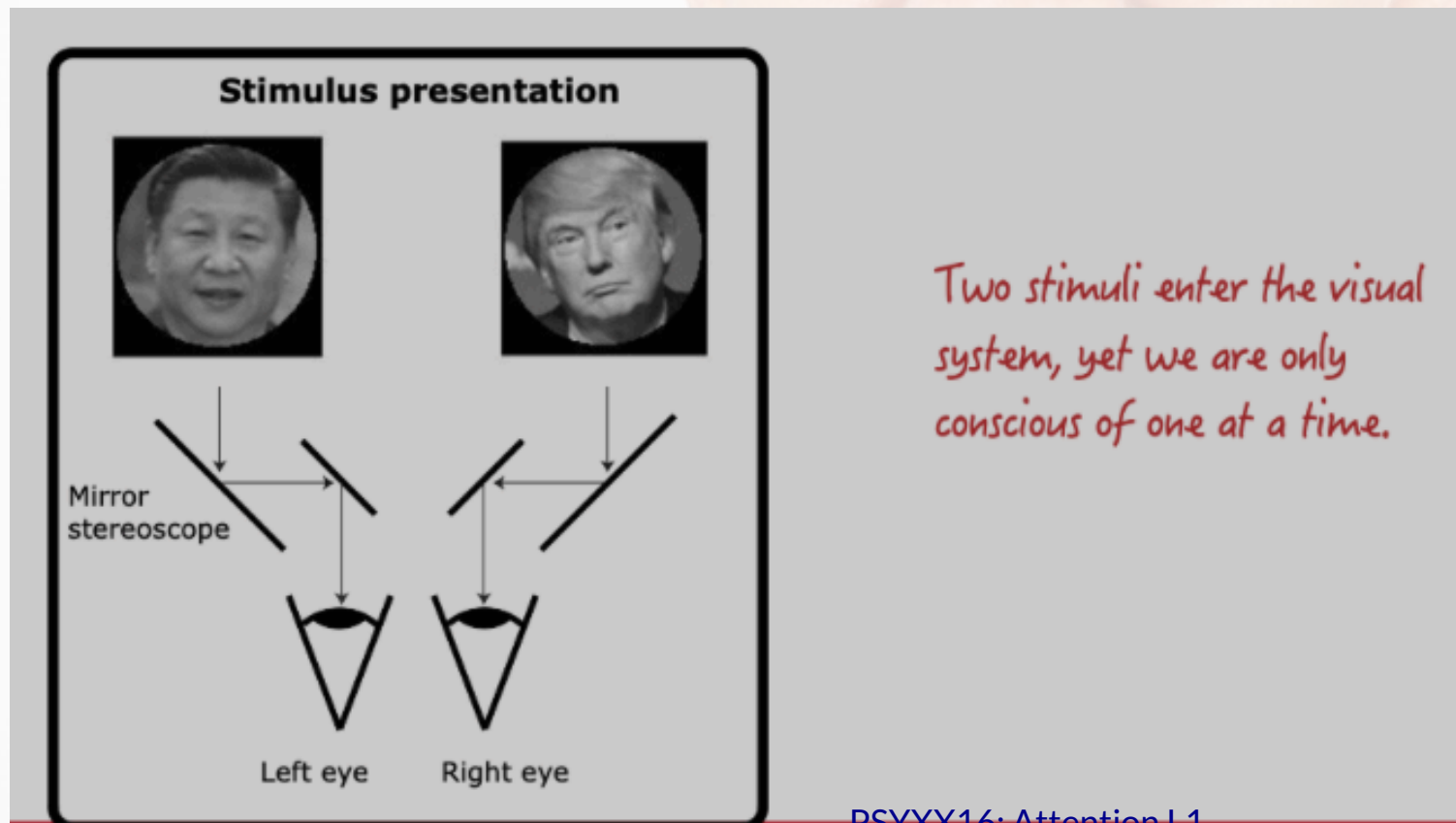


What happens when a stimulus location is selected?

- Activity makes it through the bottleneck
- Recurrent activity
- Why can't we be conscious of all stimuli?
 - Bottleneck



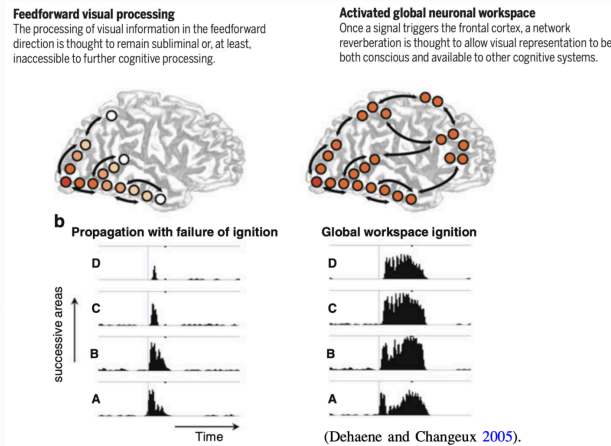
You said the two letters are processed simultaneously,
but other lecturer said the faces of Trump and Jinping
alternate



Binocular rivalry versus dual letters



Global neuronal workspace theory



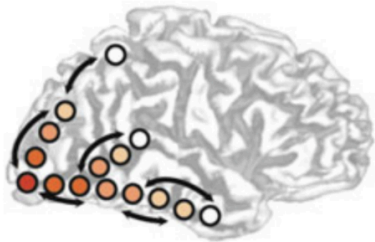
- We can't be conscious of all stimuli
- A bottleneck prior to the frontal lobe
- If a stimulus gets through, it results in reverberating widespread activity associated with that stimulus



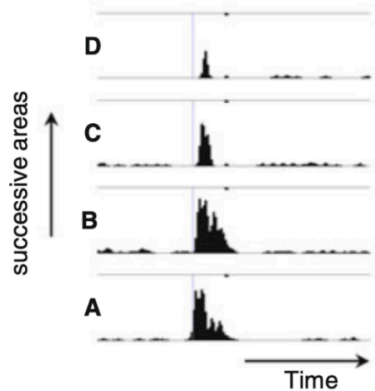
What happens when a spatial location is selected?

Feedforward visual processing

The processing of visual information in the feedforward direction is thought to remain subliminal or, at least, inaccessible to further cognitive processing.

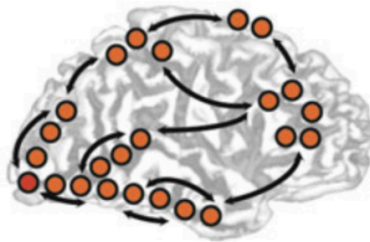


b Propagation with failure of ignition

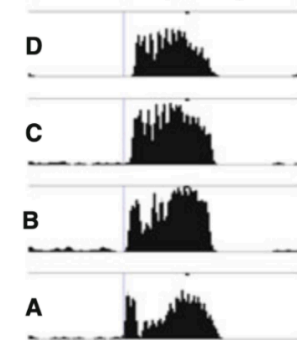


Activated global neuronal workspace

Once a signal triggers the frontal cortex, a network reverberation is thought to allow visual representation to be both conscious and available to other cognitive systems.



Global workspace ignition

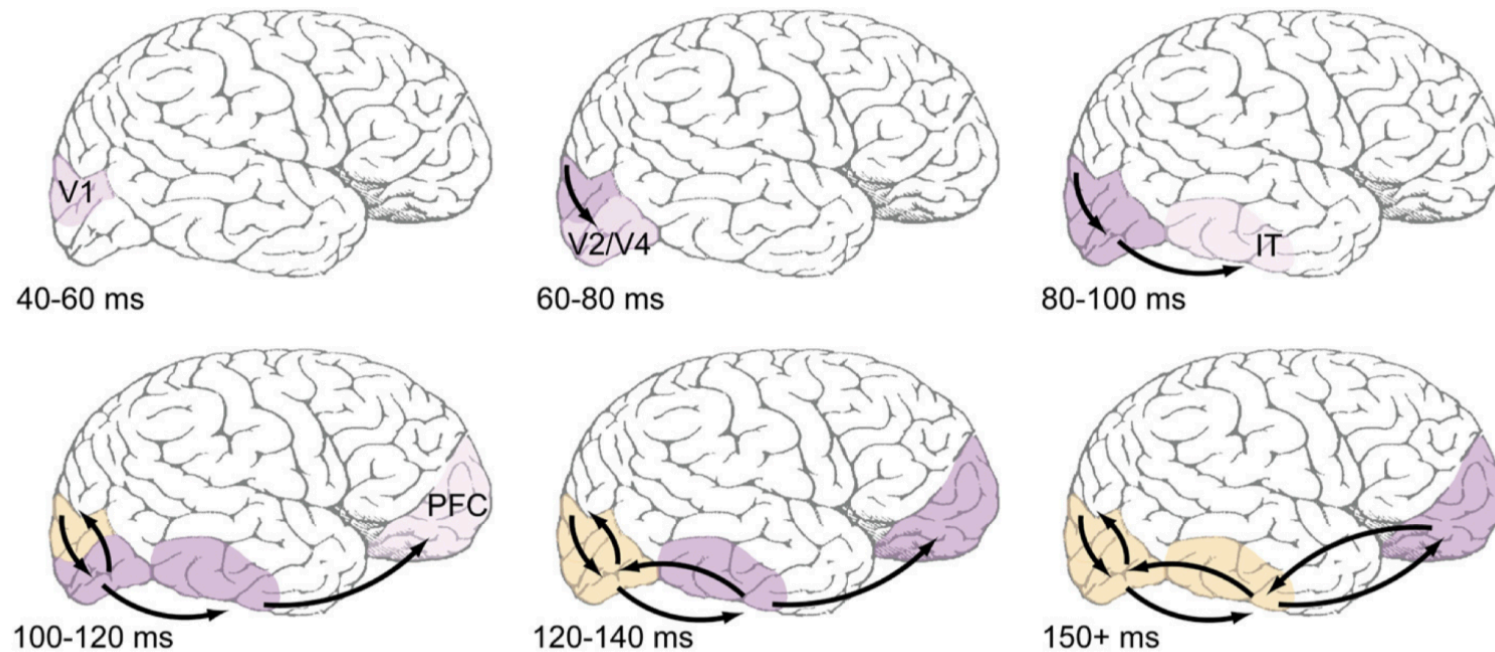


(Dehaene and Changeux 2005).



What happens when a spatial location is selected?

Consciousness is not a place but a process of widespread recurrent activity



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Thing(s) I didn't
understand

Waiting for responses ...

